C++ Course Structure

Target Audience:

- Track-A: Absolute beginners
- Track-B: Programmers with good knowledge of C

MODULE 1: Introduction to C++

(Common for both tracks – faster pace for C programmers)
Duration: 4 hours

Topics

- Evolution: C → C++ → Modern C++
- Why C++? (Systems, Embedded, Game, Finance)
- Compiling & execution process (g++, clang)
- Structure of a C++ program
- `iostream`, `namespace std`
- `cin`, `cout`, manipulators

Lab

- Simple input/output programs
- Compile using command line & IDE

MODULE 2: C++ Basics & Syntax

Topics

- Data types and variables
- Constants: `const`, `#define`
- Operators & expressions
- Control statements (`if`, `switch`, loops)
- Functions in C++
- Function overloading
- Inline functions

Lab

- Calculator program
- Menu-driven applications

MODULE 3: Memory & References (Key Difference from C)

Topics

- Pointers vs References
- `new` and `delete`
- Dynamic memory allocation in C++
- Call by value, reference, pointer
- Scope resolution operator `::`

Lab

- Dynamic array allocation
- Swapping using references

MODULE 4: Object-Oriented Programming (Core of C++)

Topics

- OOP concepts overview
- Classes and Objects
- Access specifiers
- Constructors & Destructors
- `this` pointer
- Static data members & functions
- Object lifecycle

Lab

- Student/Employee management system
- Constructor–Destructor tracing

MODULE 5: Advanced OOP Concepts

Topics

- Inheritance (single, multilevel, multiple)
- Method overriding
- Virtual functions
- Runtime polymorphism
- Base class pointers
- `final`, `override`
- Object slicing problem

Lab

- Shape hierarchy
- Polymorphic billing system

MODULE 6: Operator Overloading & Templates

Topics

- Operator overloading rules
- Unary & binary operator overloading
- Function templates
- Class templates
- Template specialization

Lab

- Complex number operations
- Generic stack implementation

MODULE 7: Exception Handling

Topics

- `try`, `catch`, `throw`
- Multiple catch blocks
- Standard exceptions
- User-defined exceptions
- Exception safety basics

Lab

- Banking transaction exception handling

MODULE 8: STL (Standard Template Library)

Topics

- STL architecture
- Containers:
  - `vector`, `list`, `deque`
  - `set`, `map`, `unordered_map`
- Iterators
- Algorithms (`sort`, `find`, `count`)
- Lambda expressions (intro)

Lab

- Word frequency counter
- Student record using `map`

MODULE 9: File Handling in C++

Topics

- File streams (`ifstream`, `ofstream`)
- Read/write text files
- Binary file handling
- Serialization basics

Lab

- File-based employee database

MODULE 10: Modern C++ Essentials (C++11/14/17 – Intro)

Topics

- `auto` keyword
- Range-based for loop
- Smart pointers (`unique_ptr`, `shared_ptr`)
- `nullptr`
- Move semantics (intro)
- RAII concept

Lab

- Memory-safe object creation using smart pointers